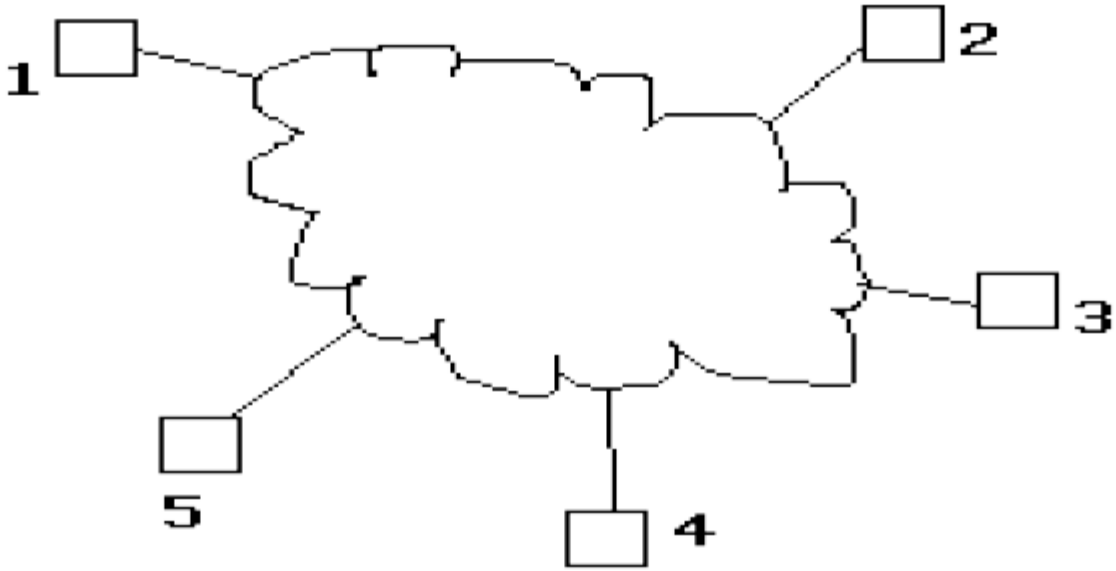
Introduction to Networks

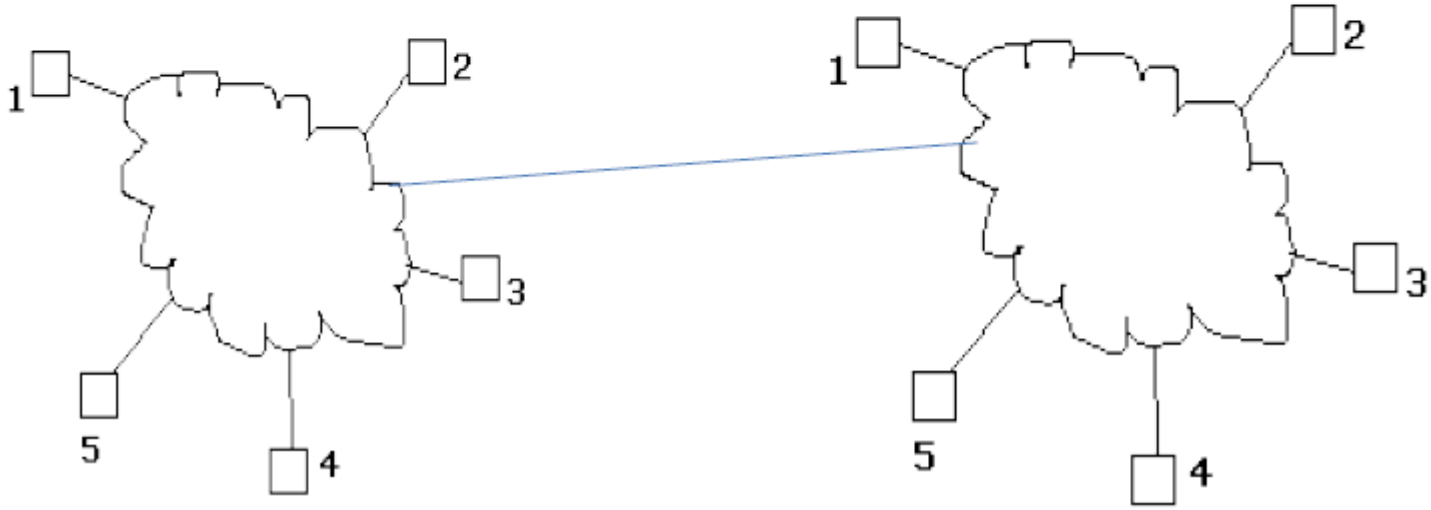
- *A Network: A group of devices that can communicate with each other over links.*
- *Each device is called a host. Each host has a unique address.*



- **Network** is a connection between two or more devices.
- Which is connected by a communication links.
- A node can be computer, printer or any other devices which is capable of sending and receiving information at each other.

INTERNET

- **An internet**: **A network of networks or connection between two or more Networks is also known as internet.** each host has an address of the form **n/h** where **n is the network number** and **h is the number of the host on network n**.



Components of Network

1) **Client** - Which gives the Request.

2) **Server** - Which gives the Response.

3) **Modems** - It Indicates Modulator / Demodulator.

4) **Router** - Which identifies the Path between Client & Server.

5) **Channels** - Which overcomes the Traffic problems.

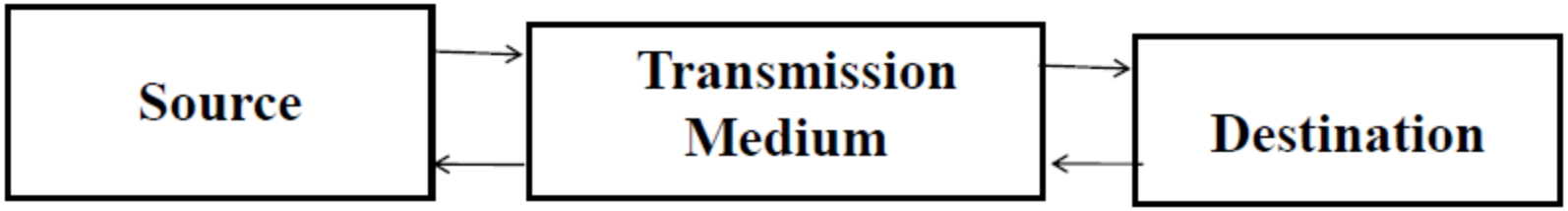
Communication Model

- Data communications are exchange of data between **two devices via some transmission medium.**
- **It should be done in two ways**

  i) Local - It takes LAN Connection.

  ii) Remote - It takes Long distance like MAN & WAN.
- Data should be Transferred in the form of **0's and 1's**

Block Diagram for Communication Model:



Characteristics of Communication Model :

1) **Delivery -** The System must deliver the data to the correct Destination.

2) **Accuracy -** The System must deliver the data at Accurate way.

3) Timeline - The System must deliver the data at **Exact Time.**

4) Jitter - It refers to the variable in the **Perfect Arrival Time.**

Components of Communication Model :

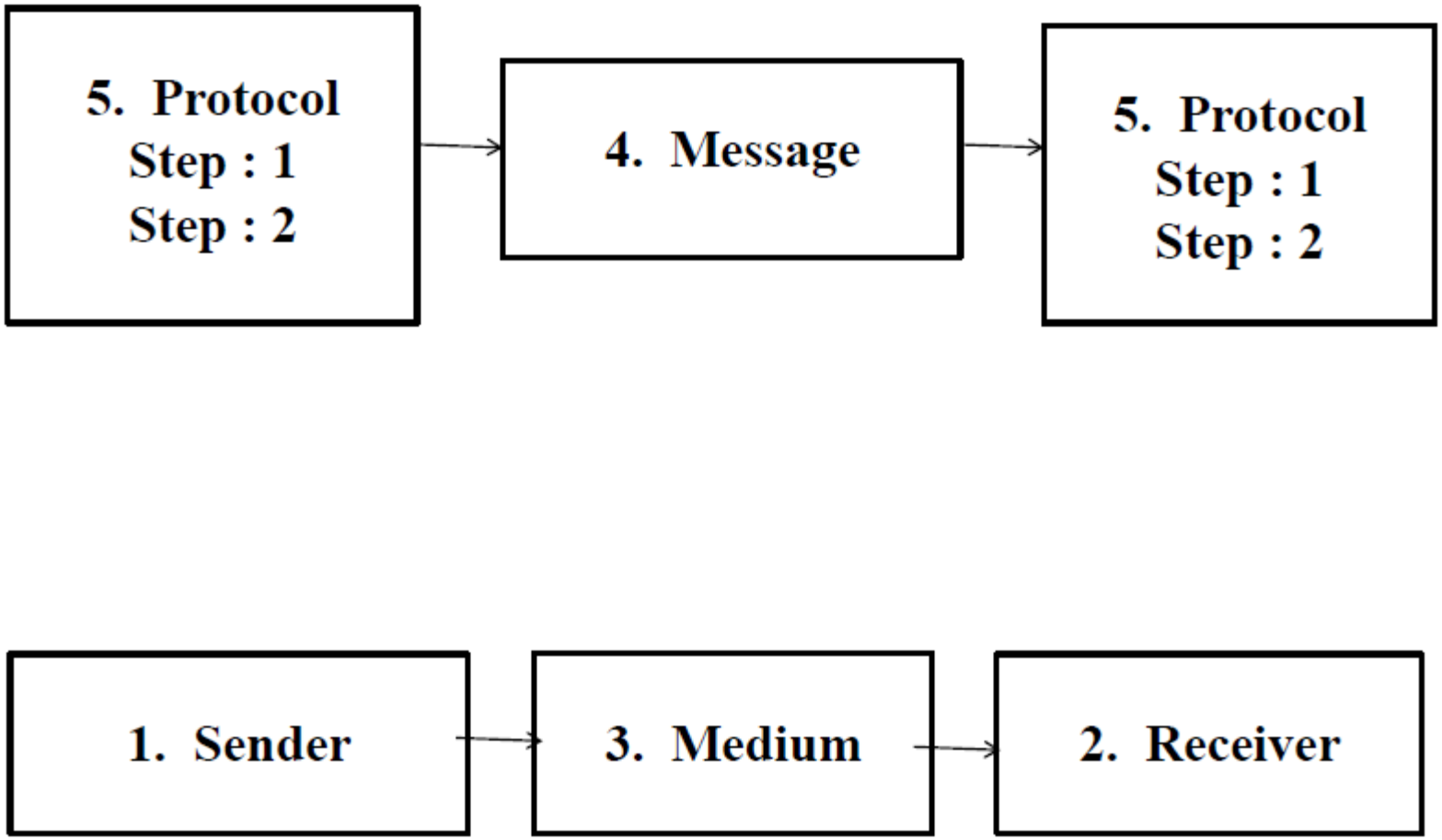
i) Sender

ii) Receiver

iii) Medium

iv) Message

v) Protocol


5. Protocol
Step : 1
Step : 2
4. Message
5. Protocol
Step : 1
Step : 2
1. Sender
3. Medium
2. Receiver

1. **Sender** : It is a device , that Sends the information to the Receiver.

2. **Receiver** : It is a device , that Receives the information from the Sender.

3. **Medium** : It is the physical path between Sender to Receiver.

4. **Message** : This is the passing Informations.

5. **Protocol** : It is a set of rules and regulations that " Governed " from data communication.

Data Transmission Concepts and Terminology

Data Transmission occurs between sender and receiver over some Transmission Medium or Transmission Media.

Transmission Media may be classified into **Two Types :**

i) **Guided Media [Wired Technology]**

ii) **Unguided Media [Wireless Technology]**

i) Guided Media (Wired Network)

- In Guided Media Signals are Passed in a " same physical path"
- Example:

  i) Twisted pair Cable

  ii) Coaxial Cable

  iii) Fiber Optic Cable

ii) Unguided Media (Wireless Network)

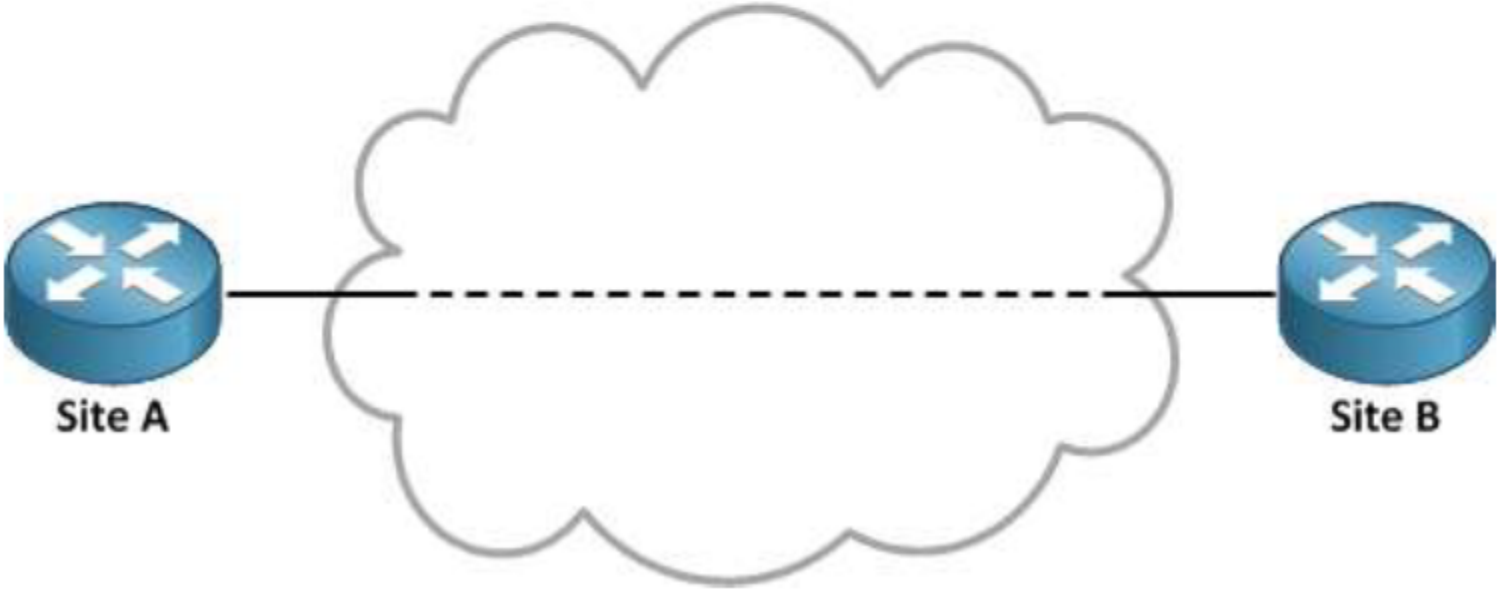
- In Unguided Media Signals are Passed in the form of " Electromagnetic Waves"
- Example :

  i) Mobile phones

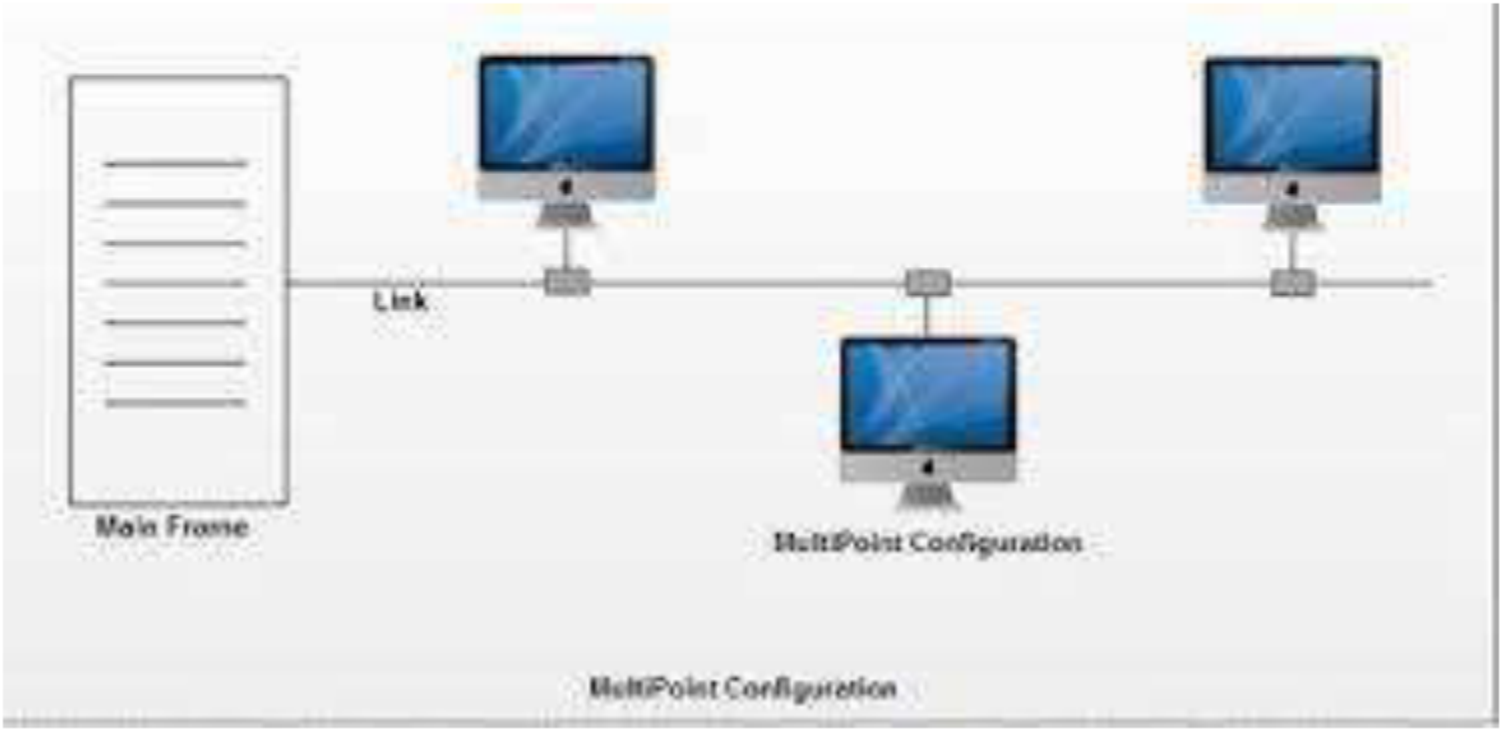
  ii) Satellite microwave

  iii) Infrared

- Point - to - Point Connection : It **Provides a dedicated links between two devices**.
- For example, a wired system that connects two computers together can be thought of a point-to-point link.



- **Multi - Point Connection : It is a link between two or more devices.** It is also known as Multi-Point configuration. The networks having multipoint configuration are called **Broadcast Networks.**



Transmission Mode

- It refers to the direction of information flow between two devices.
- Data flow is the flow of data between 2 points.
- The direction of the data flow can be described as
  - Simplex Mode
  - Half-Duplex Mode
  - Full-Duplex Mode

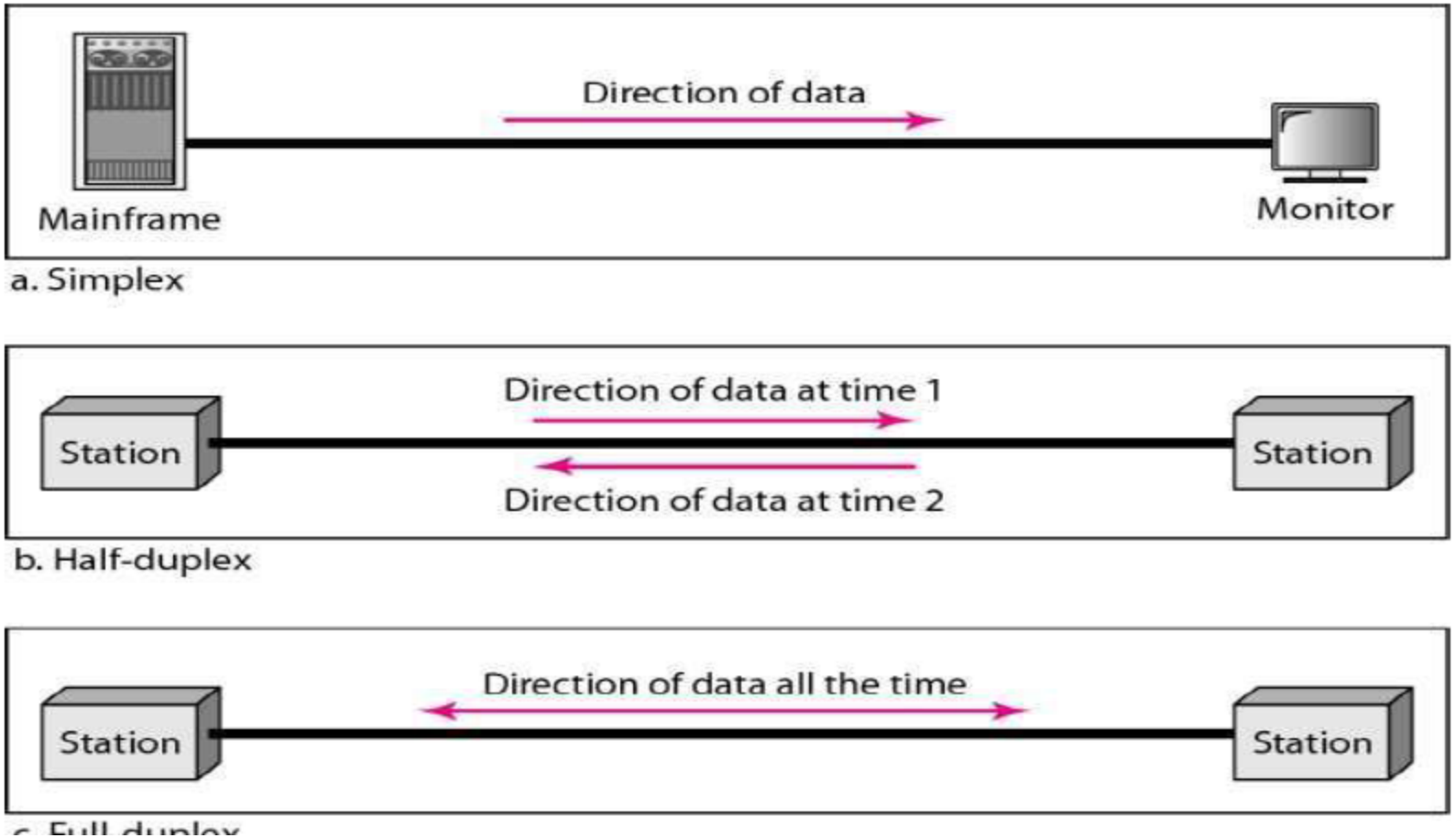
- **Simplex:** Data flows in only one direction on the data communication line (medium).

  Examples are Radio and Television broadcasts.
- **Half-Duplex:** Data flows in both directions but only one direction at a time on the data communication line.

  Ex. Conversation on walkie-talkies.
- **Full-Duplex:** Data flows in both directions simultaneously. Modems are configured to flow data in both directions.

  Ex. Phone Conversation

Data Flow


Direction of data
Mainframe
Monitor
a. Simplex
Direction of data at time 1
Station
Station
Direction of data at time 2
b. Half-duplex
Direction of data all the time
Station
Station
c. Full-duplex

Protocol

- Protocol is a set of rules that govern data communication
- It represents **what** is communicated, **when** it is communicated and **how** it is communicated.
- There are 3 key elements

✓ Syntax

✓ Semantics

✓ Timing

Syntax

- It represents **structure**, Format of data the order in which it is presented

<u>Data may contain:</u>

- First 8 bit -> Sender Address
- Second 8 bit -> Receiver Address
- Remaining bits-> message stream

SEMANTICS

- It refers the **meaning** of each section of bit

TIMING

- It refers when data sent and how fast it is sent (Says Characteristics)
- Ex:100Mbps

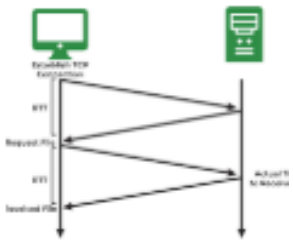
Protocol Architecture

- It is a layered structure of H/W and S/W that supports exchange of data b/w systems
- It supports distributed applications(E-Mail, File Transfer)
- Each layer of protocol architecture provides some set of rules
- There are 2 widely used protocol architecture

✓TCP/IP Architecture

✓OSI Model

throughput



Throughput is the actual data transferred over a network, calculated as Size/Effective Time, while bandwidth is the maximum theoretical capacity. Effective time includes latency (RTT) and transmission time: $\text{Effective Time} = \text{RTT} + \frac{\text{Size}}{\text{Bandwidth}}$. RTT (Round-Trip Time) causes delays in data transmission, reducing effective throughput.

Key Formulas

- **Throughput (T):** $\frac{\text{Total Data Sent (bits)}}{\text{Total Time (s)}}$
- **Effective Transfer Time (T_{eff}):** $\text{RTT} + \text{Transmission Time} = \text{RTT} + \frac{\text{Data Size}}{\text{Bandwidth}}$
- **Effective Throughput (with RTT):** $\frac{\text{Data Size}}{\text{RTT} + \frac{\text{Data Size}}{\text{Bandwidth}}}$
- **Bandwidth Delay Product (BDP):** $\text{Bandwidth (bits/s)} \times \text{RTT (s)}$

- **Propagation Time (Delay):**
  - **Definition:** Time taken for the first bit to travel from sender to receiver.
  - **Formula:** $\frac{\text{Distance}}{\text{Propagation Speed}}$.
  - **Influenced By:** Physical distance and the medium (e.g., fiber, copper).
- **Transmission Time (Delay):**
  - **Definition:** Time taken to push all bits of a packet onto the wire.
  - **Formula:** $\frac{\text{Message Size (bits)}}{\text{Bandwidth (bits/sec)}}$.

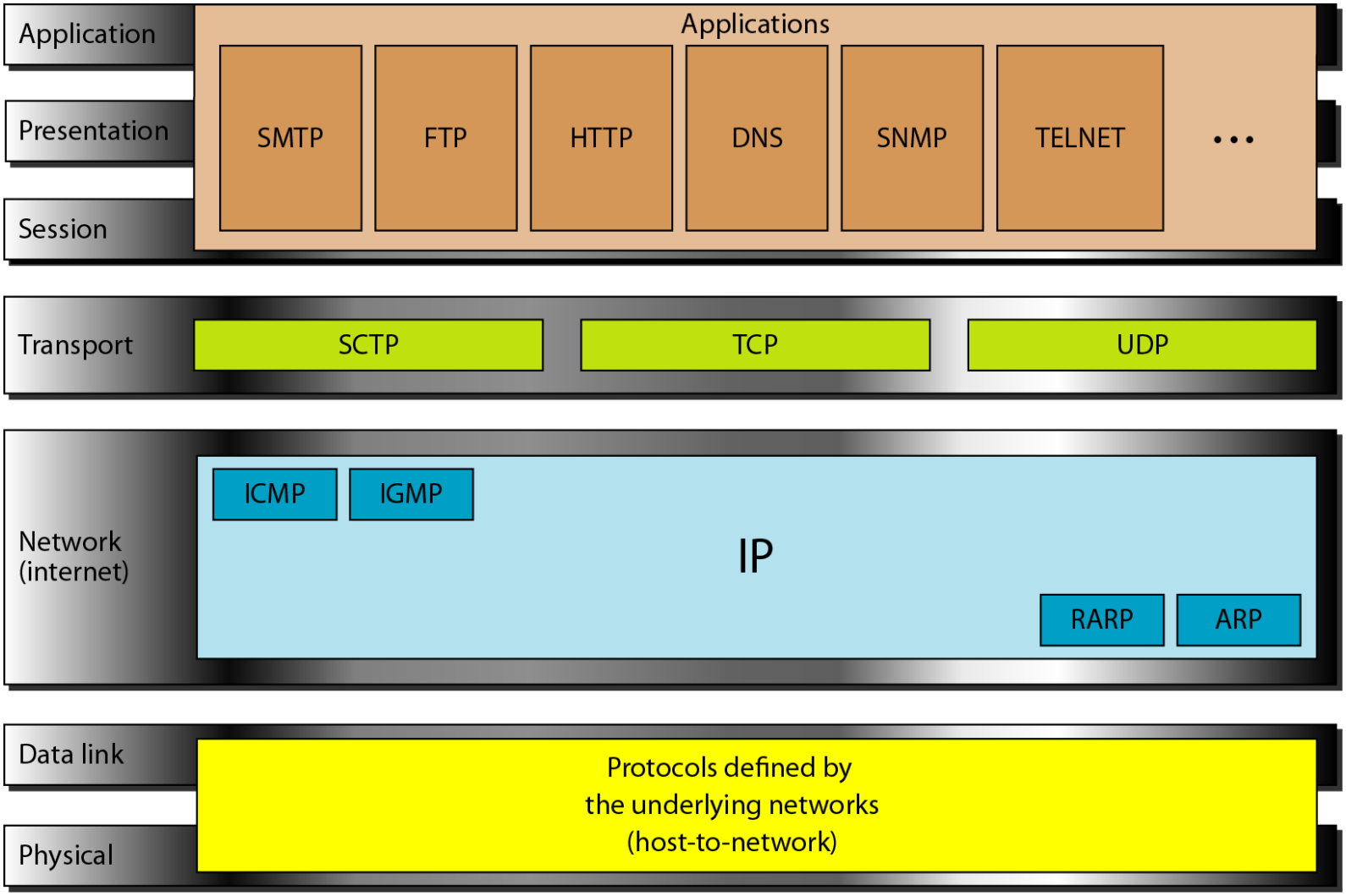
TCP/IP protocol suite

The layers in the TCP/IP protocol suite do not exactly match those in the OSI model. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.

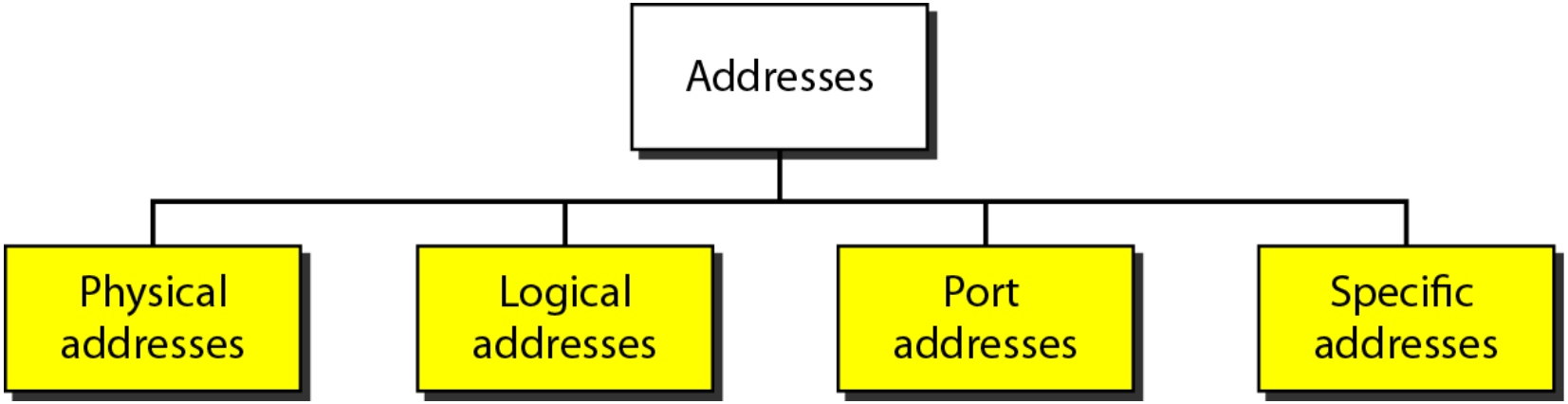
Topics discussed in this section:

Physical and Data Link Layers
Network Layer
Transport Layer
Application Layer

Figure 2.16 *TCP/IP and OSI model*

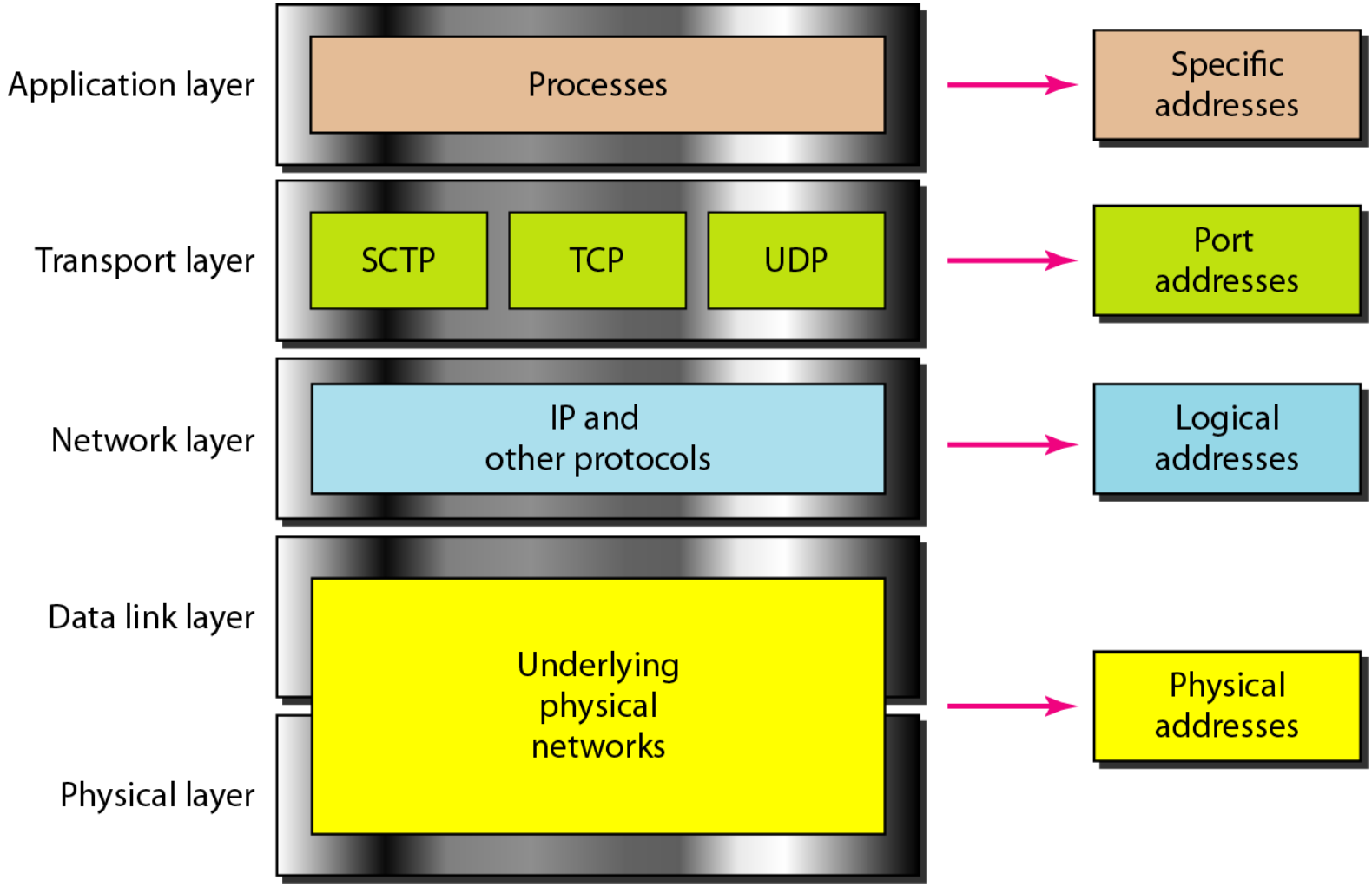


Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific.



Physical Addresses
Logical Addresses
Port Addresses
Specific Addresses

Figure 2.18 *Relationship of layers and addresses in TCP/IP*



Physical Layer

- The physical layer is responsible for carrying individual bits in a frame across the link.
- Two devices are connected by a transmission medium (cable or air).
- We need to know that the transmission medium does not carry bits; it carries electrical or optical signals.
- So the bits received in a frame from the data-link layer are transformed and sent through the transmission media, but we can think that the logical unit between two physical layers in two devices is a *bit*.

Data-link Layer

- The data-link layer is responsible for taking the datagram and moving it across the link.
- The link can be a wired LAN with a link-layer switch, a wireless LAN, a wired WAN, or a wireless WAN.
- TCP/IP does not define any specific protocol for the data-link layer
- The data-link layer takes a datagram and encapsulates it in a packet called a *frame*.

Network Layer

- The network layer is responsible for creating a connection between the source computer and the destination computer.
- The communication at the network layer is host-to-host.
- The network layer in the Internet includes the main protocol, Internet Protocol (IP), that defines the format of the packet, called a datagram at the network layer.
- IP also defines the format and the structure of addresses used in this layer.
- IP is also responsible for routing a packet from its source to its destination, which is achieved by each router forwarding the datagram to the next router in its path.

Network Layer

- IP is a connectionless protocol that provides no flow control, no error control, and no congestion control services. This means that if any of theses services is required for an application, the application should rely only on the transport-layer protocol.
- The Internet Control Message Protocol (ICMP) helps IP to report some problems when routing a packet.
- The Internet Group Management Protocol (IGMP) is another protocol that helps to send the same message to a group of recipients.
- The Dynamic Host Configuration Protocol (DHCP) helps IP to get the network-layer address for a host.
- The Address Resolution Protocol (ARP) is a protocol that helps IP to find the link-layer address of a host.

Transport Layer

- The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet (called a *segment* or a *user datagram* in different protocols) and sends it, through the logical (imaginary) connection, to the transport layer at the destination host.

Transport Layer

- Transmission Control Protocol (TCP), is a connection-oriented protocol that first establishes a logical connection between transport layers at two hosts before transferring data.
- It creates a logical pipe between two TCPs for transferring a stream of bytes.
- TCP provides flow control (matching the sending data rate of the source host with the receiving data rate of the destination host to prevent overwhelming the destination), error control (to guarantee that the segments arrive at the destination without error and resending the corrupted ones), and congestion control to reduce the loss of segments due to congestion in the network

Transport Layer

- User Datagram Protocol (UDP), is a connectionless protocol that transmits user datagrams without first creating a logical connection.
- In UDP, each user datagram is an independent entity without being related to the previous or the next one (the meaning of the term *connectionless*).
- UDP is a simple protocol that does not provide flow, error, or congestion control.
- A new protocol, Stream Control Transmission Protocol (SCTP) is designed to respond to new applications that are emerging in the multimedia.

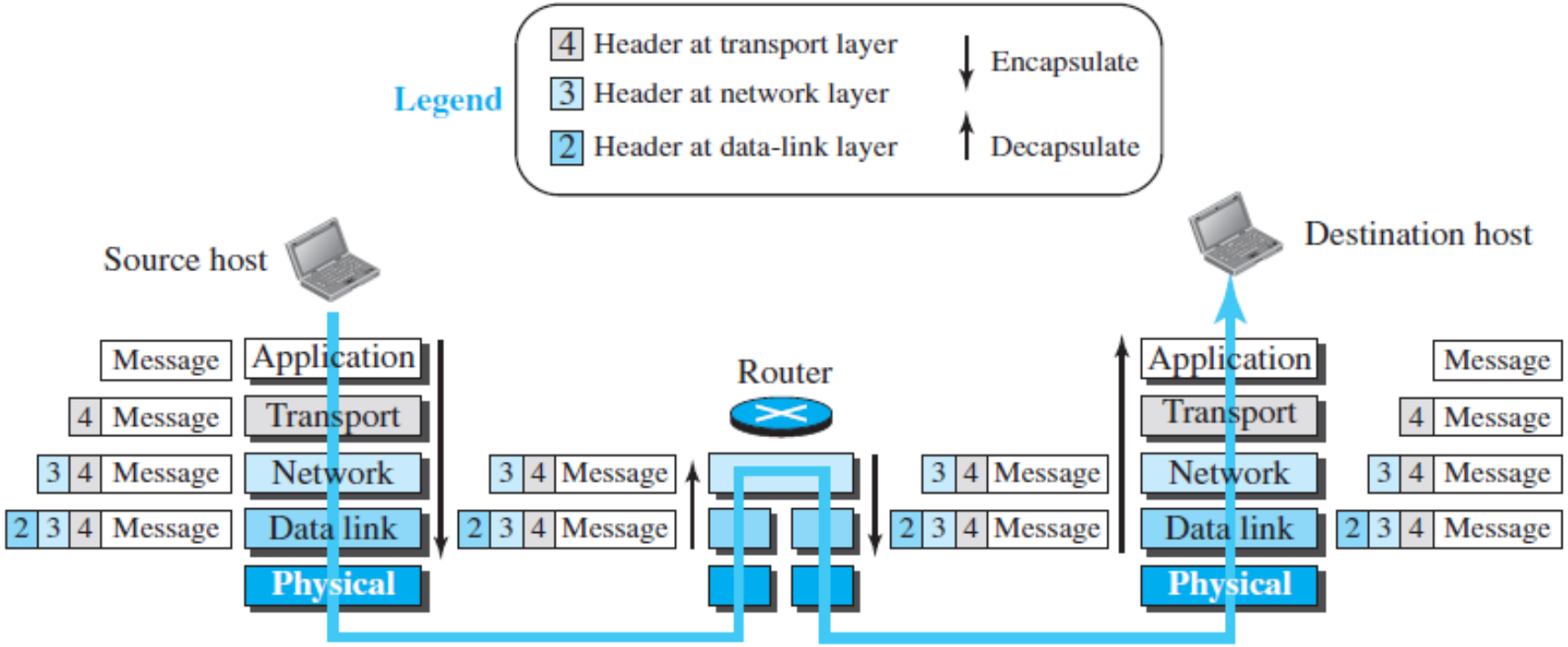
Application Layer

- Communication at the application layer is between two *processes* (two programs running at this layer).
- To communicate, a process sends a request to the other process and receives a response. Process-to-process communication is the duty of the application layer.
- The Hypertext Transfer Protocol (HTTP) is a vehicle for accessing the World Wide Web (WWW).
- The Simple Mail Transfer Protocol (SMTP) is the main protocol used in electronic mail (e-mail) service.
- The File Transfer Protocol (FTP) is used for transferring files from one host to another.

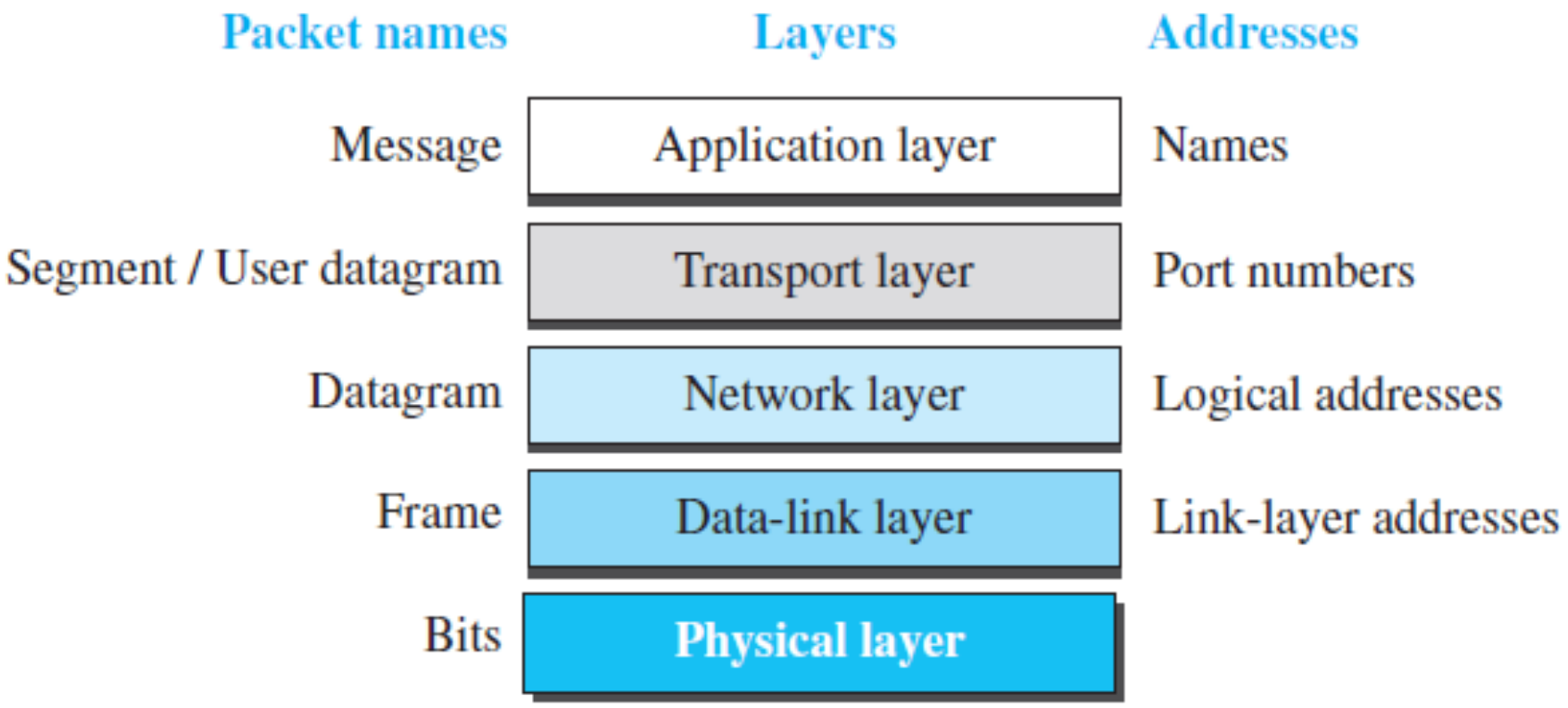
Application Layer

- The Terminal Network (TELNET) and Secure Shell (SSH) are used for accessing a site remotely. The Simple Network Management Protocol (SNMP) is used by an administrator to manage the Internet at global and local levels.
- The Domain Name System (DNS) is used by other protocols to find the network-layer address of a computer.
- The Internet Group Management Protocol (IGMP) is used to collect membership in a group.

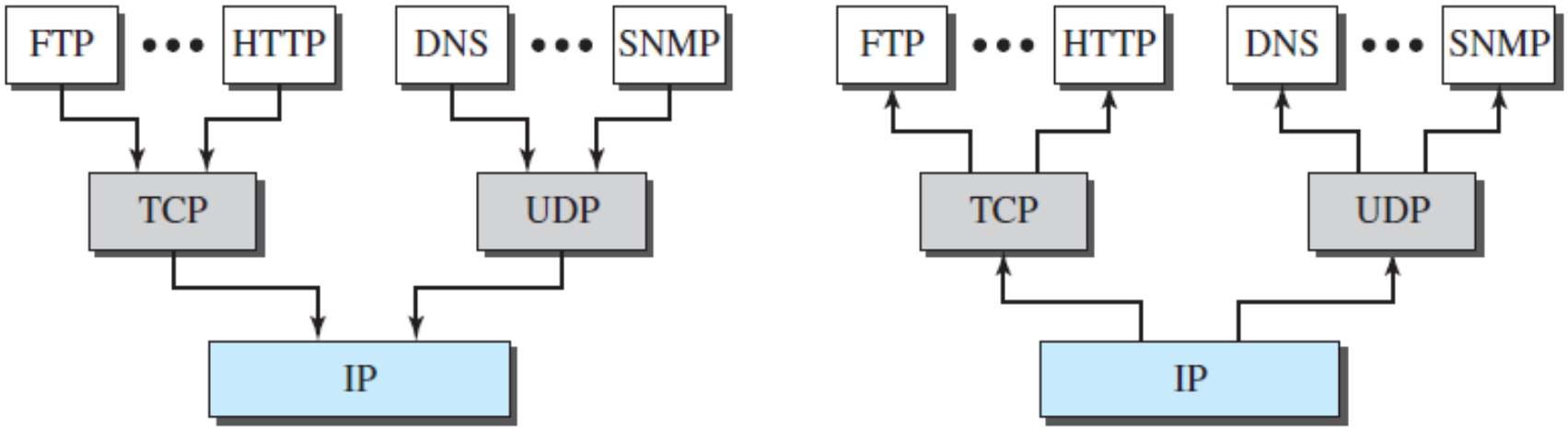
Encapsulation and Decapsulation



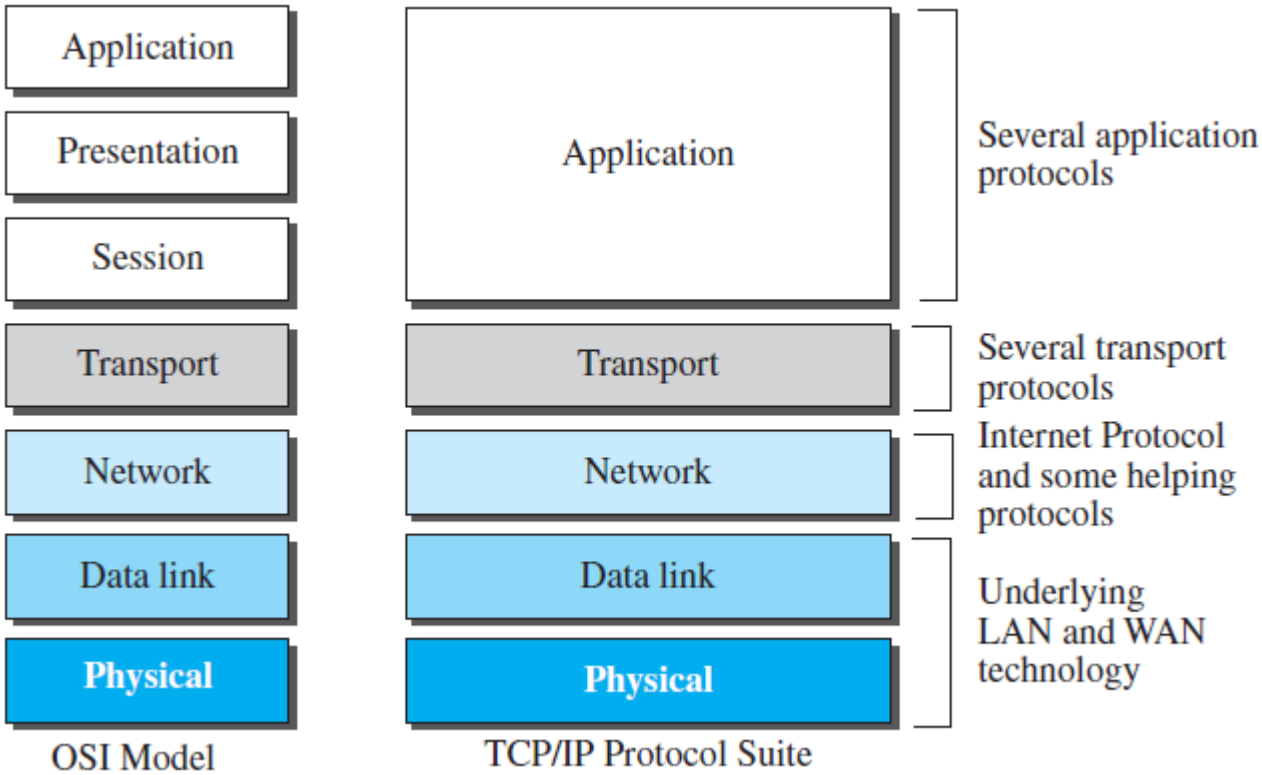
Addressing in the TCP/IP protocol suite



Multiplexing and Demultiplexing



a. Multiplexing at source

b. Demultiplexing at destination

OSI versus TCP/IP